DigitalPaani Internal Document Hub — Product Requirements Document

FIELD	VALUE
Product	DigitalPaani Internal Document Hub
Owner	Mihir Sethi (`mihir.sethi@digitalpaani.com`)
Status	v0.1 — Active development
Last updated	2026-05-15
Repository	https://github.com/mihirsethiDP/InternalTool
Live URL	https://mihirsethidp.github.io/InternalTool/

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1. Executive Summary

DigitalPaani installs and maintains instrumentation (level/pressure/DO/COD/BOD/TSS/TDS/VFD sensors, PLCs, etc.) at wastewater treatment plants. Across the project lifecycle — vendor procurement, installation, commissioning, handover, warranty, ongoing service — the company accumulates hundreds of documents per plant: vendor manuals, calibration certificates, P&IDs, I/O lists, handover packets, warranty certificates, design data sheets, and more. These are scattered across Google Drive folders with no consistent structure, no search-within-PDF, no role-based access, and no shared source of truth.

This product is a **centralised, search-first document hub** for the internal team. v0.1 ships with: full-text search across every PDF page, browse-by-filter, plant ↔ equipment ↔ sensor linking, role-gated upload with metadata, an editable taxonomy, and an architecture pre-wired for AI search.

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2. Problem Statement

Today, document handling is broken in five concrete ways:

1. **Cluttered storage.** Drive folders accumulate without conventions; locating any specific document is a hunt.
2. **No in-document search.** Search-within-PDF (e.g. "find troubleshooting steps for ERR-01 in any Brotek manual") is impossible at scale.
3. **High onboarding cost.** New joiners cannot self-serve — they must shadow senior staff to learn "where things live."

4. **Tribal knowledge.** A handful of people know where the right document is. Their unavailability blocks the rest of the team.
5. **No structured taxonomy.** Documents aren't tagged by plant / sensor / type, so even browse-by-filter doesn't work.

The qualitative cost is delayed client responses, inconsistent training, and dependency on individuals. The quantitative cost is the engineering hours spent locating documents that already exist somewhere in Drive.

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3. Goals & Non-Goals

3.1 Goals (v0.1)

#	GOAL	ACCEPTANCE SIGNAL
G1	A single URL where internal users find any document	All target document types listed in §4 are uploadable and findable
G2	Full-text search **inside PDF contents** (not just titles)	Querying a phrase that only appears in page 7 of a manual returns that document, with the matching snippet and page number
G3	Role-based access: read open to all `@digitalpaani.com` , write to allowlist	Viewer cannot upload; Uploader cannot manage users; Admin can do both
G4	Document tagging by plant, equipment, sensor model	Every document captured in v0.1 carries enough metadata to be discoverable via either browse or search
G5	First-class plant model: plant → equipment → sensor installed on equipment	The Plant detail page surfaces equipment groups, linked sensors, and plant-scope documents
G6	Architecture future-proofed for AI search	Schema includes (commented) embedding column; search provider interface allows swapping in hybrid AI search without UI changes
G7	Deployable as a static SPA on GitHub Pages with Supabase backend	Live build at the public URL with no manual server provisioning

3.2 Non-Goals (v0.1)

- Generative AI synthesis of search results (deferred to v0.2; architecture prepared).
- Automated parsing of Design Data Sheets to populate plant sensors (deferred to v0.2).

- Crawling vendor websites (Siemens, etc.) to ingest manuals (legal/ToS risk; using vendor URL fields instead).
- External-user access (clients). v0.1 is internal-only.
- Full-text extraction from Word/Excel/images (deferred — only metadata is indexed for non-PDFs in v0.1).
- Mobile-first UI (responsive but desktop-optimised).
- Versioning of documents (each upload is a discrete record).
- Granular per-document ACLs (read access is uniform within the company).

4. Personas & Use Cases

4.1 Personas

PERSONA	PRIMARY DOCUMENTS	FREQUENCY	ROLE
Electrical Engineer	Installation guides, troubleshooting steps, sensor manuals, calibration procedures	Daily	Viewer or Uploader
Domain Team	Sensor manuals, troubleshooting steps, technical data sheets	Weekly	Viewer or Uploader
Monitoring Team	Sensor manuals, troubleshooting steps, technical data sheets	Daily	Viewer
Customer Service Manager	Plant-specific documents (handover, I/O list, P&ID, warranty cert)	Daily	Viewer
Project Lead / Admin	All of the above + onboarding new joiners + managing taxonomy	Weekly	Admin

4.2 Core user stories

ID	AS A ...	I WANT TO ...	SO THAT ...
US-1	Field engineer	search "ERR-01 Brotek UT 116" and get the exact troubleshooting page	I can resolve a site issue without calling a senior
US-2	New joiner	browse all manuals for a sensor make/model the team uses	I can learn the catalog independently
US-3	CSM	open a plant page and see its handover, I/O list, P&ID, warranty cert	I can answer a client question in one screen
US-4	Project lead	upload a warranty certificate, tag it to the plant, have its contents searchable	the field team can self-serve in future
US-5	Admin	grant upload rights to a specific user	trusted teammates contribute without bottlenecking on me
US-6	Engineer	search "all level transmitters at STP Aurangabad"	I know what's installed before site visit
US-7	Project lead	add a brand-new sensor model that wasn't in our master sheet	I can document it the moment we adopt it

5. Domain Model

```
                    ┌──────────┐
                    │  Plants  │
                    └────┬─────┘
                         │ 1 ── n
              ┌──────────┼───────────────┐
              ▼          ▼               ▼
     ┌───────────┐  ┌────────────┐  ┌───────────┐
     │ Equipment │  │ Documents  │  │   PLCs    │
     └─────┬─────┘  │ (plant_id) │  └───────────┘
           │ 1      └─────┬──────┘
           │ ── n         │
           │              │ many ⇔
           │              │ document_type
           ▼              │
   ┌─────────────────┐    │
   │ PlantSensors    │    │
   │(plant_id,       │    │
   │ equipment_id,   │    │
   │ sensor_model_id)│    │
   └───────┬─────────┘    │
           │              │
           ▼              │
     ┌────────────────┐   │
     │ SensorModels   │◀──┘  (sensor_model_id)
     │  ┌─────────┐   │
     │  │  Makes  │   │
     │  └─────────┘   │
     │  ┌────────────┐│
     │  │ Categories ││
     │  └────────────┘│
     └────────────────┘
           ▲
           │ has many
           │
     ┌────────────────┐
     │ DocumentChunks │
     │ (page text +   │
     │  tsvector +    │
     │  embedding*)   │
     └────────────────┘
```

`*` = column exists in schema (commented in 001_init.sql) ready for v0.2 AI search; not used in v0.1.

5.1 Document type taxonomy & scope rules

Each document type carries a **scope** that drives required metadata on upload:

DOCUMENT TYPE	SCOPE	REQUIRED FIELDS BEYOND FILE + TITLE
Sensor Manual	`general`	Sensor model
Installation Guide	`general`	Sensor model
Troubleshooting Steps	`general`	Sensor model
Technical Data Sheet	`general`	Sensor model
Test Certificate	`plant_sensor`	Plant + Equipment + Sensor model
Calibration Certificate	`plant_sensor`	Plant + Equipment + Sensor model
Handover Document	`plant`	Plant
I/O List	`plant`	Plant
P&ID	`plant`	Plant
Onboarding Document	`plant`	Plant
Design Data Sheet	`plant`	Plant
Warranty Certificate	`plant_with_sensor_refs`	Plant (sensors referenced via PDF text → FTS)

Admins can add new document types from the Admin panel; the scope defaults to `general` and can be edited in the DB.

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6. Functional Requirements

6.1 Authentication & Roles

- **FR-AUTH-1** Sign-in is via Supabase magic link (email OTP).
- **FR-AUTH-2** A `profiles` row is auto-created on first sign-in. Default role: `viewer` .
- **FR-AUTH-3** Three roles: `viewer` , `uploader` , `admin` .
- **FR-AUTH-4** Role transitions are admin-only.
- **FR-AUTH-5** Sign-out clears the session and routes to `/login` .
- **FR-AUTH-6** Supabase URL/Redirect config must allow the deployed origin (`https://mihirsethidp.github.io/InternalTool/`).
- **FR-AUTH-7** *Future:* email-domain restriction to `@digitalpaani.com` via DB trigger or auth hook.

6.2 Search (the heart of the product)

- **FR-SEARCH-1** A single hero search input on `/` (the landing page) accepts free text and queries all indexed document chunks.
- **FR-SEARCH-2** Results show: document title, type, plant (if any), equipment (if any), sensor make + model (if any), matched page number, and a highlighted snippet from the matching chunk.
- **FR-SEARCH-3** Ranking uses Postgres `ts_rank` over a tsvector generated from each chunk's text. Document title / plant name / sensor model fallback matches with low rank.
- **FR-SEARCH-4** Clicking a result opens the document — either at the matched page (via `#page=N` on the signed URL) or the vendor URL if no file is stored.
- **FR-SEARCH-5** Example query chips reveal possible search idioms (Hick's law — keep visible set ≤ 5).
- **FR-SEARCH-6** Search is reactive (no Search button required; updates as user types).
- **FR-SEARCH-7** *Future (v0.2):* hybrid AI search — semantic similarity over embeddings + FTS via Reciprocal Rank Fusion + Claude synthesises an answer with citations. UI surface is identical.

6.3 Browse

- **FR-BROWSE-1** `/browse` lists **all documents** by default — no filter required.
- **FR-BROWSE-2** An always-visible search input filters by document text.
- **FR-BROWSE-3** Filters: Document type, Plant, Sensor category, Make, Model.
- **FR-BROWSE-4** Filters are fully **independent** — selecting one does not require any other.
- **FR-BROWSE-5 Progressive disclosure:** Sensor category / Make / Model only appear when the chosen document type is sensor-relevant (sensor manual, install guide, troubleshooting, datasheet, test cert, calibration cert, warranty cert).
- **FR-BROWSE-6 Cascade within a group:** Model dropdown only appears after a Make is picked; lists models of that make.

6.4 Plants

- **FR-PLANT-1** `/plants` lists all plants in card layout with name + location.
- **FR-PLANT-2 Admin only:** an inline "+ New plant" button opens an inline form.
- **FR-PLANT-3** `/plants/:id` shows a plant hero (name, location, stats), an Equipment section, a Sensors section, and a Documents section.
- **FR-PLANT-4 Equipment** is a first-class entity owned by a plant. Uploader/admin can add and remove equipment inline.
- **FR-PLANT-5** Sensors installed on the plant are shown as **cards grouped by equipment** (with an Equipment dropdown to filter to one group). Sensors not linked to any equipment appear under "Unassigned."

- **FR-PLANT-6** Linking a sensor to a plant requires: Make (selected first) → Model (cascading) → Equipment (optional).
- **FR-PLANT-7** The Documents section supports a Document-type filter dropdown.
- **FR-PLANT-8** A prominent "+ Upload document" button in the plant header opens the global Upload modal pre-filled with the current plant.

6.5 Sensor catalog

- **FR-SENSOR-1** `/sensors` lists all sensor models grouped by category in cards.
- **FR-SENSOR-2** Search input + filters: Category, Make, Model.
- **FR-SENSOR-3** Make/Model cascade: model list filters to selected make.
- **FR-SENSOR-4** **Uploader/admin** can click "+ New sensor" → opens AddSensorModal with Make (with datalist autocomplete), Category, Model number, optional Description, optional Vendor URL.
- **FR-SENSOR-5** Make is **find-or-create** — typing a new make creates it; typing an existing one (case-insensitive match) reuses it.
- **FR-SENSOR-6** `/sensors/:id` shows the model's specs, suitability, technical details, vendor URL, and all linked documents.
- **FR-SENSOR-7** Pagination at 24 per page on the catalog list.

6.6 Upload

- **FR-UPLOAD-1** A global "+ Upload" button in the header opens a modal. Available to **uploader** and **admin** only.
- **FR-UPLOAD-2** The modal accepts: PDF, Word (.doc/.docx), Excel (.xls/.xlsx), images (.png/.jpg/.jpeg/.gif/.webp).
- **FR-UPLOAD-3** Drag-and-drop file picker with hover/active styling and a file-type emoji indicator after pick.
- **FR-UPLOAD-4** Title field **auto-fills** from the file name on file pick. User can edit; auto-fill stops once user edits.
- **FR-UPLOAD-5** Document-type dropdown plus "Recent types" chips (top 3 most-recently used in the last 30 uploads). Same pattern for "Recent plants" and "Recent makes."
- **FR-UPLOAD-6** Fields shown adapt to the chosen document type's scope (see §5.1):
  - `general` → Make + Model required, plant ignored
  - `plant` → Plant required
  - `plant_sensor` → Plant + Equipment + Sensor model required
  - `plant_with_sensor_refs` → Plant required (sensors via FTS)
- **FR-UPLOAD-7** Inline link "+ Can't find it? Add new sensor" opens AddSensorModal without leaving the upload flow; the new model is auto-selected on return.

- **FR-UPLOAD-8** On submit: file uploads to Supabase Storage, a `documents` row is inserted, and (PDFs only) text is extracted client-side via pdf.js, chunked, and inserted into `document_chunks` for search.
- **FR-UPLOAD-9** Progress bar visible; non-PDFs are still indexed by title chunk so they're findable by metadata.
- **FR-UPLOAD-10** A plant page's "+ Upload document" button pre-fills the plant. Same for sensor model detail.

6.7 Admin

- **FR-ADMIN-1** `/admin` is admin-only.
- **FR-ADMIN-2 Users panel:** lists all users, allows changing role between viewer/uploader/admin.
- **FR-ADMIN-3 Plants panel:** inline create.
- **FR-ADMIN-4 Document types panel:** inline create with key (snake_case) + label + sort_order.
- **FR-ADMIN-5** *Future:* Makes panel (rename/merge/delete makes); audit log; deletion of documents.

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7. Non-Functional Requirements

7.1 Security

- **NFR-SEC-1** All tables have Row-Level Security enabled.
- **NFR-SEC-2** Read policies require `auth.role() = 'authenticated'`.
- **NFR-SEC-3** Write policies require `current_role() IN ('uploader','admin')`. Profile updates require `current_role() = 'admin'`.
- **NFR-SEC-4** Storage bucket `documents` is private; access via Supabase-signed URLs with 10-minute TTL.
- **NFR-SEC-5** No secrets in the frontend — anon key is public-by-design; service_role key is never used client-side.
- **NFR-SEC-6** Supabase project URL and anon key are injected at build time via GitHub Actions secrets.
- **NFR-SEC-7** *Open item:* email-domain enforcement (`@digitalpaani.com`) — currently relies on social discipline; to be hardened via Supabase auth hook before broader rollout.

7.2 Performance

- **NFR-PERF-1** Search RPC must return ≤ 50 hits within 500 ms for catalogs up to 10,000 documents and 200,000 chunks (limited by Supabase tier).
- **NFR-PERF-2** Initial page render ≤ 2 s on a typical broadband connection (LCP target 1.8 s).
- **NFR-PERF-3** PDF text extraction at upload time happens in-browser; up to 100-page PDFs should complete within 10 s on modern laptops.

7.3 Reliability

- **NFR-REL-1** Single-region Supabase Postgres + Storage; daily backups (Supabase default on Pro tier).
- **NFR-REL-2** Deployments are atomic via GitHub Pages; rollback by re-running a prior workflow.
- **NFR-REL-3** RPC + DB migrations are versioned in `supabase/migrations/` ; re-runnable safely (idempotent where reasonable).

7.4 Accessibility & UX

- **NFR-UX-1** WCAG AA contrast on brand colour (`#193458`) against white surfaces.
- **NFR-UX-2** Keyboard-accessible navigation through all primary actions.
- **NFR-UX-3** UX design follows: **Hick's law** (limit visible choices in suggestion chips), **Miller's law** (paginate at 24, recent chips capped at 3), **Aesthetic-Usability Effect** (consistent visual system across all pages), **Law of Proximity** (related fields grouped, generous whitespace between groups), **Law of Similarity** (consistent badges, buttons, cards, table styles).

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8. Information Architecture

```
/                         Search (Home)
/login                    Magic-link sign-in (unauthenticated)
/browse                   All documents with filters + in-page search
/plants                   Plant directory; admin-only "+ New plant"
/plants/:id               Equipment + sensors + plant-scope documents
/sensors                  Sensor catalog grouped by category
/sensors/:id              Model detail + linked documents
/admin                    Admin-only: users, plants, document types
```

Routing is hash-based (`HashRouter`) for GitHub Pages compatibility.

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9. Technical Architecture

9.1 Stack

- **Frontend:** React 18 + TypeScript + Vite + Tailwind CSS, hash routing, TanStack Query for data.
- **PDF handling:** `pdfjs-dist` (worker-backed, in-browser text extraction).
- **Backend:** Supabase (Postgres + Auth + Storage + RLS + RPC).
- **Hosting:** GitHub Pages (static); GitHub Actions builds and deploys on push to `main`.

9.2 Database (key tables)

TABLE	PURPOSE
`profiles`	One row per user; carries `role`.
`sensor_categories`	6 clean families (Flow, Level, Pressure, Water Quality, Maintenance & Safety, Other).
`sensor_makes`	Vendor names; deduped case-insensitively.
`sensor_models`	(make, category, model_no) tuples + descriptive fields.
`plcs`	PLC catalog (parallel to sensors).
`plants`	Operational sites.
`equipment`	Children of `plants` (air blowers, tanks, pumps, etc.).
`plant_sensors`	(plant, equipment, sensor_model) — sensors installed on equipment.
`document_types`	Configurable taxonomy with `scope`.
`documents`	One row per uploaded file; links to plant / equipment / sensor_model.
`document_chunks`	Page-level text chunks; carries `tsvector` (used today) and `embedding vector(1536)` *(reserved for v0.2).*

9.3 Search RPC

`public.search_documents(q, p_plant_id, p_sensor_model_id, p_plc_id, p_type_key, p_category_id, p_make_id, p_equipment_id, p_limit)` — performs ts_query ranking over `document_chunks.tsv` with metadata joins; returns hits with snippets via `ts_headline`. Title/plant/sensor fallback matches included.

When AI search is enabled, this RPC is superseded by `hybrid_search_documents` accepting an additional `embedding vector(1536)` parameter; the frontend swap point is `src/lib/search/index.ts` (`SearchMode` env var).

9.4 Migrations (chronological)

FILE	PURPOSE
`001_init.sql`	Initial schema, RLS, FTS, search RPC, storage bucket, document-type seed.
`002_seed.sql`	Original catalog seed from master xlsx (retained for reversibility).
`003_simplify_catalog.sql`	Option B: wipe noisy models/PLCs; collapse 68 categories to 6; dedupe makes.
`004_equipment_and_scope.sql`	`equipment` table, `documents.equipment_id` , `plant_sensors.equipment_id` , `document_types.scope` , extended search RPC.
`005_demo_seed.sql`	4 makes / 4 models / 1 demo plant + equipment + 17 dummy docs with searchable chunks.

9.5 Deployment

- Build: GitHub Actions on push to `main` . Secrets: `VITE_SUPABASE_URL` , `VITE_SUPABASE_ANON_KEY` .
- Output: `dist/` artefact deployed to GitHub Pages (`/InternalTool/` base path).
- Local dev: `npm run dev` with `.env.local` .

10. UX Principles Applied

PRINCIPLE	WHERE IT SHOWS UP
Hick's law (limit choices)	Search chips capped at 5; Recent chips capped at 3.
Miller's law (chunks of ~7)	Sensor catalog paginates at 24; recent chips at 3.
Aesthetic-Usability Effect	Shared design system (cards, badges, buttons, gradient headers). Search page is the hero.
Law of Proximity	Upload modal groups related fields ("Link this document" cluster). Plant detail groups sensors by equipment with whitespace between groups.
Law of Similarity	All page headers use one component; all primary buttons share gradient style; all badges share shape and palette.
Progressive disclosure	Browse only shows Make/Model when relevant; Make → Model cascade; "+ Add new sensor" inline link in upload.

11. Out of Scope (v0.1)

- AI search synthesis (architecture ready, not enabled).
- DDS auto-parse to populate plant sensors.
- Vendor-website ingestion / web scraping (legal/ToS risk).
- Word/Excel/image text extraction at upload (only metadata indexed).
- Document versioning.
- Per-document or per-folder ACLs beyond role.
- External (client) access.
- Mobile-first redesign.
- Bulk import / bulk reassignment tools.
- In-app PDF viewer (delegating to browser PDF viewer with `#page=N`).
- Audit log of edits.

12. Roadmap

v0.2 — Smart search (target ~2–4 weeks)

- Turn on `pgvector`; backfill `document_chunks.embedding` for existing rows.
- Implement `hybridAIProvider` — embed query, hybrid rank, Claude synthesis with citations.
- "AI answer" card above the results list on `/` and `/browse`.

v0.3 — Workflow & ingestion

- DDS parser: when a Design Data Sheet is uploaded, extract the sensor/equipment list and offer to bulk-link.
- Word / Excel / image OCR text extraction at upload (`mammoth`, `xlsx`, Tesseract / Vision API).
- Admin → Makes panel: merge / rename / delete.

v0.4 — Lifecycle & governance

- Document versioning (replace + retain history).
- Audit log (who uploaded / edited / deleted what, when).
- Soft-delete + admin restore.
- Email-domain enforcement at sign-in.
- Per-document expiry reminders (e.g. warranty cert nearing expiry).

v1.0 — Beyond docs

- External access for select clients on selected plants (read-only, share-link based).
- Mobile-optimised view.
- Cross-plant analytics: "which sensors fail most often" derived from troubleshooting doc clusters.

13. Success Metrics

METRIC	DEFINITION	V0.1 TARGET
TTD — Time to Document	Median seconds from search query to opening the right document	< 30 s
Coverage	% of "actively referenced" documents ingested into the hub vs. still in Drive	80% within 4 weeks of internal rollout
Self-serve rate	% of internal document requests resolved without escalation	> 75% within 8 weeks
Active uploaders	Distinct users uploading per week	≥ 5
Search recall	For a curated set of 20 known phrases, % returning the expected document in the top 5	≥ 90%
Drive abandonment	New documents created in legacy Drive folder vs. uploaded to hub	< 10% by week 8

14. Risks & Mitigations

#	RISK	LIKELIHOOD	IMPACT	MITIGATION
R1	Search relevance is poor for sensor jargon (FTS alone is keyword-bound)	Medium	High	v0.2 hybrid AI search adds semantic similarity. Today: rely on rich chunking + title fallback matches.
R2	Adoption stalls — team keeps using Drive	High	High	Make upload modal frictionless (drag-drop, recent chips, auto-title). Communicate clear cutover dates. Track Drive abandonment metric.
R3	Storage cost grows unpredictably as PDFs accumulate	Medium	Medium	Supabase Pro Storage is ~$0.021/GB/mo; even 100 GB ≈ $2.10/mo. Monitor monthly.
R4	Email rate limit on magic links blocks first-time users	Medium	Low	Dashboard send-link bypass documented; eventually move to a paid SMTP provider for higher limits.
R5	A non-admin gains upload role via social engineering	Low	Medium	Admin role changes are audited via Postgres logs; future audit-log feature.
R6	A future schema change breaks the search RPC for older clients	Low	Medium	RPC is versioned in migrations; only one in-use at a time; clients refetch on deploy.
R7	GitHub Pages static-site limits become a problem (no custom auth headers, hash routing)	Low	Medium	If hit, lift-and-shift to Vercel or Cloudflare Pages with zero frontend changes.
R8	DDS auto-parse (v0.3) is technically harder than expected (formats vary per template)	Medium	Low	Acceptable to keep manual linking flow as fallback.

15. Open Questions

- **OQ-1** Should warranty certificates be modelled as many-to-many with sensor models (structured) or kept text-search-only? *Current decision: text-only — revisit if filtering "warranty certs covering Brotek UT 116" by metadata becomes a real workflow.*
- **OQ-2** Should we allow non-admins to add new makes (via the find-or-create flow) or restrict make creation to admins? *Current: any uploader can implicitly create makes — pragmatic for*

v0.1, may tighten if clutter returns.

- **OQ-3** How do we present the same document being relevant to multiple plants (e.g. one generic Brotek manual)? *Current: store once with `sensor_model_id`; surface on every plant that installs the sensor. Adequate.*
- **OQ-4** Should expired warranty certs auto-archive? *Out of scope for v0.1; v0.4 candidate.*

16. Glossary

TERM	DEFINITION
Hub	This product — the centralised document store.
Plant	An operational wastewater treatment site (e.g. "STP Aurangabad").
Equipment	A physical asset within a plant (e.g. aeration tank, filter feed pump).
Sensor model	A specific make + model combination in the catalog (e.g. Brotek UT 116).
Make	The manufacturer/vendor of a sensor or PLC (e.g. Siemens).
Category	Sensor family — Flow, Level, Pressure, Water Quality, Maintenance & Safety, Other.
Document type	The kind of document (manual, install guide, calibration cert, etc.) with an associated `scope`.
Scope	Determines which metadata fields are required on upload: `general`, `plant`, `plant_sensor`, `plant_with_sensor_refs`.
Chunk	A unit of indexed text — one row per (document, page) in `document_chunks`.
FTS	Postgres Full-Text Search via `tsvector` + `ts_rank`.
RLS	Row-Level Security — Postgres-native authorisation rules.
Magic link	Email-based passwordless sign-in flow.
Hybrid search	Combination of keyword (FTS) and semantic (vector cosine) ranking — planned for v0.2.

Appendix A — Visual / UX inventory of pages

PAGE	HERO ELEMENT	KEY ACTIONS	SECTIONS
Search (`/`)	Centred bold heading + large search input	Type, click chip	Recently uploaded grid
Browse	Gradient hero with result count stat	Search, filter	Filter card + results
Plants	Gradient hero with plant count	"+ New plant" (admin)	Plant cards
Plant detail	Gradient hero with stats	"+ Upload document"	Equipment, Sensors, Documents
Sensors	Gradient hero with stats	"+ New sensor" (uploader+)	Filter card, category groups
Sensor detail	Gradient hero	"+ Upload document"	Specs / Suitability / Technical / Vendor, Documents
Admin	Gradient hero	n/a	Users, Plants, Document types
Login	Logo + branded card	Sign in	Magic-link form

End of document — see the repo for source of truth and `supabase/migrations/` for the authoritative schema.